

French Standard

PERMANENT RAILWAY EQUIPMENT Wide hexagon nuts, grade B – HL nuts

F: Ecrous hexagonaux larges, grade B – Ecrous HL
D: Ortsfeste Bahnanlagen – Hohe Sechskantmuttern, Produktklasse B – HL
Muttern

French Standard qualified further to a decision by the Director General of the Afnor on 5 June 1990, effective from 5 July 1990.

Replaces the registered standard of November 1979

correspondence

At the date of publication of this standard, there are no other known European or international standards on the subject.

analysis

This standard defines the methods for defining, manufacturing and inspecting wide hexagonal nuts (HL nuts) used in permanent railway equipment.

key words

International Technical Thesaurus: Permanent railway equipment, track, rail, fastener, hexagonal nut, manufacture, characteristic, dimension, acceptance inspection.

updates

With regard to the previous edition:

- application of tolerances for threaded fasteners per standard NF E 27-021;
- setting up a control plan;
- brought into line with current standards.

corrections

The French and original document only is :

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Wide hexagon nuts, grade B

NF F 50-007

HL nuts

July 1990

1 SCOPE AND PURPOSE

This document governs the supply of wide hexagon nuts used in permanent railway equipment, especially for fastening the rails onto the concrete sleepers (sleeper screws – cam bolts – rail clip bolts – fishplate bolts). Appendix A “Single Sampling Plan” is an integral part of the standard.

These nuts are designed to be tightened using the 38 mm wrenches of standard NF F 76-014.

2 REFERENCES

NF A 03-115	General delivery conditions for steel products.
NF A 55-101	Zinc ingots.
NF A 91-102	Metal coatings – Electrolytic deposit of zinc and cadmium on iron and steel
NF A 91-111	Metal coatings – Thickness measurement – Coulometric method by anode corrosion.
NF A 91-112	Metal coatings – Thickness measurement using magnetic methods – General Specifications.
NF A 91-113	Metal coatings – Thickness measurement using the Eddy Current ¹ method - General Specifications.
NF A 91-121	Galvanization by immersion in molten zinc – Finished iron, steel and cast iron products.
A 91-122	Metal coatings – Finished hot galvanized steel products – Recommendations regarding the design and utilization of galvanized products.
NF A 91-472	Chromatising treatment of zinc or cadmium electrolytic deposits – Specifications and Test Method.
NF E 03-001	Triangular thread metric threading – ISO thread profile
NF E 03-053	Triangular thread metric threading – ISO threading tolerance system – Average quality 6 H/6 g for fasteners and other standard applications.
NF E 03-151	Triangular thread metric threading – Threading inspection using limit gauges – General considerations.
NF E 03-152	Triangular thread metric threading - Threading inspection using limit gauges – External thread gauges.
NF E 03-153	Triangular thread metric threading - Threading inspection using limit gauges – Internal thread gauges – Part tolerances per NF E 03-051.
NF E 03-154	Triangular thread metric threading - Threading inspection using limit gauges – Plain screw and internal thread gauges
NF E 25-003	Fasteners – General classification criteria.
NF E 25-005	Fasteners – Test Method.
NF E 25-007	Fasteners – Ordering and Delivery Conditions
NF E 25-009	Fasteners – Electrolytic coating of threaded components.
NF E 25-400	Fasteners – Group standard for nuts.
NF E 27-021	Fasteners – Tolerances for threaded bolts – Diameters 1.6 to 150 mm.
NF E 27-711	Controlled tightness fasteners for metal structures – Hexagon head bolts – Dimensions and tolerances.
NF F 76-014	Permanent railway equipment – 38 mm open-ended wrenches for track bolts.

¹ “Courant de Foucault” in French.

3 DESIGNATION

Nuts compliant with this standard are designated as follows (in order):

- designation "HL nut";
- symbol "M" followed by the nominal diameter;
- type 2 per standard NF E 25-003.
- quality class;
- reference of this standard;
- protective coating (if any).

For example: HL Nut, M 22, class 5, NF F 50-007, Zn 12-D-Fe

NB: Only the indications: "nuts, HL shape symbol, quality class and reference of this standard" shall be indicated to designate a nut in the case of the designation of a railway bolt using this type of nut.

4 MATERIAL - MANUFACTURE

4.1 Material

The steel grade to be used for manufacturing nuts shall be left to the discretion of the supplier who shall be entirely responsible for this choice.

The steel grades used must allow all the mechanical characteristics indicated in chapter 5.3 to be obtained on the finished parts.

4.2 Manufacture

HL nuts are manufactured exclusively by hot forging. The internal threading may be machined or formed. It must comply with the ISO thread profile per standard NF E 03-001.

No machining shall be performed after protective treatment, where applicable. When hot galvanization is required however, the internal threading shall be made after coating.

4.3 Anti-corrosion treatment

The nuts may be delivered black (rough condition) or protected by electrolytic galvanization or hot galvanization in accordance with standards:

- NF A 91-102, NF A 91-472 and NF E 25-009 for electrogalvanizing.
- NF A 91-121 and A 91-122 for hot galvanizing.

In the case of galvanization, the minimum quantity of zinc used shall be type Z 6 in accordance with standard NF A 55-101 and the weight of the zinc deposit shall be greater than or equal to 200 g/m².

In the case of electrogalvanization, the protection shall be type Zn12/D/Fe. Other protective coatings may be used with the customer's approval.

5 CHARACTERISTICS

5.1 Physical characteristics

The nuts shall be as forged, internally sound with no calamine inclusions. The acceptance limits for these flaws are given in chapter 5 of standard NF E 25-400, type 2.

5.2 Mechanical characteristics

The HL nuts shall be class 5 unless otherwise specified in the order.

Their mechanical characteristics shall be set by standard NF E 25-400.

5.3 Dimensional characteristics -Weights

5.3.1 Dimensions and weights

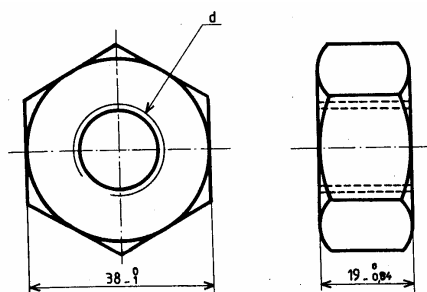


Figure 1

TABLE 1

Nominal diameter d	Weight (in kg) *
18	0.155
20	0.146
22	0.138
24	0.129

* The weights are calculated using a specific density of 7.85 kg/dm³ for steel. They are given as an indication only.

15.3.2 Tolerances

The manufacturing tolerances applied are those of grade B of standard NF E 27-021.

The internal threading shall be quality:

- 6H, for nuts delivered "black" and "electrogalvanized" in accordance with standard NF E 03-053;
- 6 AZ, for galvanized nuts in accordance with standard NF E 27-711.

6 SAMPLING

The type of tests, acceptable quality level (AQL) and control level are given in table 2.

The sampling plan is shown in appendix A.

TABLE 2

Tests	AQL	Type 2	
		Nature	Control Level Single Plan
Appearance inspection (unacceptable surface flaws)	1	0	S 2
Dimensional and shape characteristics			
Dimension of the drive device	2.5	0	S 3
Threaded GO gauge	2.5	0	S 4
Threaded NO GO gauge	2.5	0	S 4
Height	2.5	0	S 3
Shape defects	2.5	0	S 3
Perpendicularity of threading to bearing face	2.5	0	S 3
Mechanical and functional characteristics			
Minimum hardness.	1.5	0	S 2
Maximum hardness	1.5	0	S 2
Proof test	0.65	0	S 2
Crushing test	1.5	0	S 2
Anti-corrosion coating	1.5	0	S 2
* O = systematic inspection of characteristics			

7 INSPECTION

7.1 Inspection of material

The inspection of the raw material used does not necessarily have to be submitted to the customer's quality department. The supplier has the ability to procure materials that have been subjected to a specific inspection performed by the production plant; they shall however be delivered accompanied by a factory product inspection certificate, in accordance with the requirements of paragraph 5.3.2 of standard NF A 03-115.

7.2 Inspection of nut threading

Nuts delivered black or electrogalvanized:

- the threading shall be inspected using max-min plug gauges in accordance with standards NF E 03-151 to NF E 03-154.

Hot galvanized nuts:

- the threading, made after galvanization, shall be inspected using a max-min gauge in accordance with the provisions of standard NF E 27-711.

7.3 Inspection of mechanical characteristics

The mechanical characteristics inspections are performed following the methods described in standard NF E 25-005.

7.4 Inspection of anti-corrosion protection

The quality of the anti-corrosion treatment shall be checked in accordance with standards:

- NF A 91-102 – NF A 91-111 – NF A 91-113
- NF A 91-121 – NF A 91-472
- A 91-122

NB: Type C or D passivation shall be used on electrogalvanized products in accordance with the order requirements. If unspecified in the order, type D passivation shall be used.

7.5 Acceptance

The nuts are normally presented for acceptance in delivery condition, that is after application of an anti-corrosion protection if any.

It is however possible for acceptance to take place before protective treatment, if this is performed in a different plant to that of the supplier with whom the order has been placed. The latter shall nevertheless retain full responsibility for the quality of the supply delivered. It is up to this supplier to make all the necessary provisions for having the quality of the protective treatment inspected.

The release date for acceptance by the customer's quality department shall be notified in writing by the management of the production plant or its authorized representative.

This notification shall indicate opposite the product name, the number of parts presented in each batch as well as the references of the order to which the parts correspond.

On delivery, a certificate attesting that the production and inspection conditions have been observed and specifying the results, shall be handed over to the customer's production inspector.

The serial tests may be performed in the absence of the customer railway network's representative, unless otherwise specified in the order or attached documents. In all cases, the test results shall be communicated or kept at the disposal of the latter.

7.6 Batches

HL nuts shall be delivered grouped into batches. Each batch shall only contain parts of the same type and same dimensions, manufactured under the same conditions.

7.7 Test type and frequency

HL nuts shall be subjected to verifications and tests by statistical sampling in accordance with table 2 or any other method approved by the customer.

7.8 Results

Any non-conforming test results shall entail in the refusal of the corresponding batch.

However, only when the non-conformity concerns the dimensional characteristics or appearance of the nuts, the supplier shall be able to resubmit the batch concerned after sorting it item by item.

8 MARKING

HL screws shall be marked with the manufacturer's identification symbol, in accordance with the requirements of standard NF E 25-400.

9 CONTROL AND SURVEILLANCE OF PRODUCTION QUALITY

The supplier shall implement an organization, methods and means that enable it to ensure the quality of its products and the inspection of the supplies intended for the customer railway network.

The customer railway network shall ensure by means of specific auditing, surveillance or inspection actions performed at the supplier's works, that the provisions made are effective.

The way in which the customer railway network shall exercise this control according to its own regulations shall be specified in the order.

10 TRACEABILITY

The supplier shall establish and operate a system for the identification and traceability of the products and components at all stages of production, inspection and delivery.

It shall ensure the traceability of all operations, products or constituent parts of products contracted to other suppliers, in order to be able to easily recall all past data concerning them, to link them together and to attribute them unequivocally to the products concerned.

Traceability shall concern the following points in particular:

- origin of the material, manufacturing method and contractual characteristics;
- processing operations and treatments;
- dimensional inspections.

Traceability shall enable all items having undergone the same production and inspection operations to be retraced.

11 DELIVERY

The order shall indicate the type of packaging required for each type of supply, as well as the quantity of items per package. The packaging used for shipping must not have been previously used for another purpose which could cause any damage to the parts to be packaged.

Unless otherwise indicated in the order, each package shall carry a securely attached label, clearly and indelibly marked with the following indications in accordance with standard NF E 25-007:

- supplier's name or mark;
- order number;
- type and quantity of the packaged parts;
- reference to the product standard;
- type of coating if any.

NB: The supplier may add any other information necessary to product traceability.

12 WARRANTY

The HL nuts shall be guaranteed by the supplier for two years, under normal conditions of use, against any manufacturing defects that were not detected on acceptance at the factory.

Any nuts that, during the warranty period, are found to contain manufacturing defects rendering them unsuitable for use, or that are liable to reduce their service life, shall be scrapped.

The scrapped nuts shall be replaced at the supplier's expense within one month of receipt of the notice sent to the supplier.

The products scrapped shall be held at the supplier's disposal with a view to their replacement.

13 STORAGE

Independently of any other regulations, the provisions (documents, delivery times, precautions, etc.) for the storage of data providing evidence of the compliance of the products with this standard, and with the particular conditions of the order or its attached documents, are defined by the customer railway network which shall notify the supplier of them. All the necessary precautions shall be taken to prevent such documents from being damaged during the required storage time.

Appendix A
(being an integral part of the standard
SINGLE SAMPLING PLAN

A.1 SAMPLING TABLE

QTY OF ITEMS IN THE BATCH TO BE CHECKED								Accepted	AQL					LQ ₁₀ when AQL is:				
Special control levels				Control Levels				Refused										
S1	S2	S3	S4	I	II	III			0.65	1	1.5	2.5	4	0.65	1	1.5	2.5	4
2	2	2	2	2	2	2	Sampled	A R	20	13	8	5	3	11	16	25	37	54
500	150	50	25	50	15	8	Qty defective		0	0	0	0	0					
501	151	51	26	51	16	9	Sampled	A R	20	13	8	5	3	11	16	25	37	54
35.000	1200	150	90	90	25	15	Qty defective		0	0	0	0	0					
35.001	1201	151	91	91	26	16	Sampled	A R	20	13	8	5	3	11	16	25	37	27
•	35.000	500	150	150	50	25	Qty defective		0	0	0	0	1					
	35.001	501	151	151	51	26	Sampled	A R	20	13	8	20	13	11	16	25	18	27
	•	3200	500	280	90	50	Qty defective		0	0	0	1	1					
		3201	501	281	91	51	Sampled	A R	20	13	32	20	20	11	16	12	18	25
		35.000	1200	500	150	90	Qty defective		0	0	1	1	2					
		35.001	1201	501	151	91	Sampled	A R	20	50	32	32	32	11	7.6	12	16	20
		500.000	10.000	1200	280	150	Qty defective		0	1	1	2	3					
		500.001	10.001	1201	281	151	Sampled	A R	80	50	50	50	50	4.8	7.6	10	13	18
		•	35.000	3200	500	280	Qty defective		1	1	2	3	5					
			35.001	3201	501	281	Sampled	A R	80	80	80	80	80	4.8	6.5	8.2	11	14
			500.000	10.000	1200	500	Qty defective		1	2	3	5	7					
			500.001	10.001	1201	501	Sampled	A R	125	125	125	125		4.3	5.4	7.4	9.4	12
			•	35.000	3200	1200	Qty defective		2	3	5	7	10					
				35.001	3201	1201	Sampled	A R	200	200	200	200	200	3.5	4.6	5.9	7.7	10
				150.000	10000	3200	Qty defective		3	5	7	10	14					
				150.001	10001	3201	Sampled	A R	315	315	315	315	315	2.9	3.7	4.9	6.4	9.0
				500.000	35000	10000	Qty defective		5	7	10	14	21					
				500.001	35.001	10001	Sampled	A R	500	500	500	500	315	2.4	3.1	4.0	5.6	9.0
				•	150.000	35000	Qty defective		7	10	14	21	21					
					150.001	35001	Sampled	A R	800	800	800	500	315	1.9	2.5	3.5	5.6	9.0
					500.000	150000	Qty defective		10	14	21	21	21					
					500.001	150001	Sampled	A R	1250	1250	800	500	315	1.8	2.3	3.5	5.6	9.0
					•	500.000	Qty defective		14	21	21	21	21					
						500001	Sampled	A R	2000	1250	800	500	315	1.4	2.3	3.5	5.6	9.0
							Qty defective		21	21	21	21	21					

A.2 EXAMPLE OF HOW TO USE THE TABLE FOR A SINGLE SAMPLING PLAN

The control sheet defines the following parameters:
Control level, type of plan (single or double), AQL

Example	Level 3	Single plan	AQL: 1
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SINGLE PLAN

The inspector reads off the table: 5,000 items – quantity of parts to be sampled
- decision to be taken (A: accepted/R: refused)

CONTROL PLAN

Number defective parts = 0
Batch accepted

Batch of 5,000 parts 13 parts sampled

Number of defective parts ≥ 1
Batch refused

The customer has less than 10% probability of accepting batches for which the proportion of defective parts is greater than or equal to 16%

LQ 16